

Stony Brook University
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Fall 2016

FINAL EXAM

NAME:

SBU ID #:

Instructions: Answer all 20 multiple choice questions. Fill out your answers in the provided **scantron**. Do not forget to put your name and ID number in the scantron sheet.

This exam is **open books/notes**. Calculators are allowed but other electronic devices are **not** allowed.

GOOD LUCK!

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1. Which of these utility functions represent the same preferences as $U(x, y) = \sqrt{xy}$?

- A. $U(x, y) = 10\sqrt{xy}$.
- B. $U(x, y) = xy$.
- C. $U(x, y) = \ln(x) + \ln(y)$
- D. All of the above represent the same preferences.

ANSWER: D

2. For the utility function $U(x, y) = 4x + 2y$, the bundle (3, 2) provides the same utility as the bundle

- A. (2, 3).
- B. (2, 4).
- C. (2, 5).
- D. (3, 3).

ANSWER: B

3. If the price of x falls, the budget constraint

- A. shifts outward in a parallel fashion.
- B. shifts inward in a parallel fashion.
- C. rotates outward about the x-intercept.
- D. rotates outward about the y-intercept.

ANSWER: D

4. If utility is given by $U(x, y) = x^2 + y^2$ and $p_x = 2$, $p_y = 3$, $I = 50$, this person will choose

- A. (10, 10).
- B. (15, 6.67).
- C. (25, 0).
- D. (0, 50/3).

ANSWER: C

5. If an individual's utility function for coffee (x) and cream (y) is given by $U(x, y) = \min\{x, 5y\}$, the demand function for coffee is given by

- A. $x = \frac{I}{2p_x}$.
- B. $x = I(p_x + p_y)$.
- C. $x = \frac{I}{p_x + 0.2p_y}$.
- D. $x = \frac{I}{(p_x + p_y)^2}$.

ANSWER: C

6. Assume x and y are the only two goods a person consumes. If after a rise in p_x the quantity demanded of y increases, one could say

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- A. the income effect dominates the substitution effect.
- B. the substitution effect dominates the income effect.
- C. it is still impossible to determine whether the substitution or income effect dominates.
- D. none of the answers is correct.

ANSWER: B

7. If an individual buys only two goods and these must be used in a fixed relationship with one another (e.g., coffee and cream for a coffee drinker who never varies the amount of cream used in each cup), then

- A. there is no substitution effect from a change in the price of coffee.
- B. there is no income effect from a change in the price of coffee.
- C. Giffen's Paradox must occur if both coffee and cream are inferior goods.
- D. an increase in income will not affect cream purchases.

ANSWER: A

8. Which of the following demand functions is *not* homogeneous of degree zero in p_x , p_y and I ?

- A. $x = \frac{I}{p_x + p_y}$.
- B. $x = I^{0.5} p_x^{-0.3} p_y^{-0.2}$.
- C. $x = I \frac{p_y}{p_x^2}$.
- D. $x = I \frac{p_y}{p_x}$.

ANSWER: D

9. If goods x and y are complements, then the cross price elasticity of demand between them will be

- A. positive.
- B. negative.
- C. zero.
- D. infinity.

ANSWER: B

10. If an individual's supply of labor curve is "backward bending", then

- A. the substitution effect always dominates the income effect.
- B. the income effect always dominates the substitution effect.
- C. the substitution effect dominates at low real wage levels and the income effect dominates at high real wage levels.
- D. the income effect dominates at low real wage levels and the substitution effect dominates at high real wage levels.

ANSWER: C

11. When an individual's wage rises, the income effect tends to

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- A. increase hours worked.
- B. decrease hours worked.
- C. leave hours worked unchanged.
- D. it is impossible to predict what will happen to hours worked.

ANSWER: B

12. Suppose an individual has a fixed amount of wealth to allocate between consumption in two periods (c_1 and c_2). Any funds not spent in period 1 will earn interest (at the rate r) which will increase purchasing power in period 2. Consider four possible reactions to an increase in r :

- I. c_1 increases.
- II. c_1 decreases.
- III. c_2 increases.
- IV. c_2 decreases.

Which of these is consistent with the hypothesis that both c_1 and c_2 are normal goods?

- A. I, II, III, and IV.
- B. I, II, and IV, but not III.
- C. I, III, and IV, but not II.
- D. I, II and III, but not IV.

ANSWER: D

13. Suppose the production function for good q is given by $q = 3k + 2l$ where k and l are capital and labor inputs. Consider three statements about this function:

- I. The function exhibits constant returns to scale.
- II. The function exhibits diminishing marginal productivities to all inputs.
- III. The function has a constant rate of technical substitution.

Which of these statements is true?

- A. All of them.
- B. None of them.
- C. I and II but not III.
- D. I and III but not II.

ANSWER: D

14. Which of the following production functions exhibits a constant elasticity of substitution?

- A. $q = 3k + 2l$.
- B. $q = k^{0.5}l^{0.5}$.
- C. $q = 1 - \frac{1}{k} - \frac{1}{l}$.
- D. All of the above have a constant elasticity of substitution.

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ANSWER: D

15. For the cost function $C = q^{0.8}v^{0.4}w^{0.6}$ consider the following statements:

- I. The function exhibits decreasing average cost.
 - II. The function is homogeneous of degree 1 in v and w .
 - III. The elasticity of marginal cost with respect to v exceeds the elasticity with respect to w .
- A. None is true.
 - B. All are true.
 - C. Only I is true.
 - D. Only I and II are true.

ANSWER: D

16. An increase in the wage rate will have a greater effect on average costs

- A. the larger the proportion labor costs are of total costs and the easier it is to substitute capital for labor.
- B. the larger the proportion labor costs are of total costs and the harder it is to substitute capital for labor.
- C. the greater is the diminishing marginal product of labor.
- D. the greater are returns to scale.

ANSWER: B

17. If a price-taking firm's production function is given by $q = \sqrt{l}$, its supply function is given by

- A. $q = 2pw$.
- B. $q = \frac{p}{2w}$.
- C. $q = \frac{2p}{w}$.
- D. $q = \frac{2w}{p}$.

ANSWER: B

18. A profit-maximizing firm's demand function for labor can be found by differentiating

- A. the profit function with respect to w .
- B. the cost function with respect to w .
- C. the supply function with respect to w .
- D. the production function with respect to w .

ANSWER: A

19. If the market for bottled spring water is characterized by a very elastic supply curve and a very inelastic demand curve, an outward shift in the supply curve would be reflected primarily in the form of

- A. higher prices.

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- B. higher output.
- C. lower prices.
- D. lower output.

ANSWER: C

20. Suppose that the price elasticity of demand for a product is -1 and that the price elasticity of supply is $+1$. Assume also that the income elasticity of demand is $+2$. Then an increase in income of 10% will raise equilibrium price by

- A. 10%.
- B. 5%.
- C. 20%.
- D. an amount that cannot be determined.

ANSWER: A